

Generalized Polynomial Chaos for Non-intrusive Uncertainty Quantification in Computational Fluid Dynamics



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Introduction

Complex aerodynamic analysis and design of aircraft use high-fidelity computational fluid dynamics (CFD) tools for shape optimization, for example, whereby some robustness is achieved by considering uncertain operational, environmental, or manufacturing parameters. Cruise flight conditions are generally transonic, as the flow may become locally supersonic depending on the wing profiles. Hence, high-fidelity simulations must be carried out in order to obtain a detailed description of the flow structure for optimization purposes. When it comes to consider variable parameters for sensitivity and robustness analyses, non-intrusive methods for uncertainty quantification (UQ) are typically considered in CFD. Indeed, the complex flow solvers are preferably treated as black boxes in order to compute the output quantities of interest that are required to evaluate the objective function of an optimization process. The latter is often expressed in terms of moments of the quantities of interest, such as the mean, standard deviation, or even higher-order moments (skewness, kurtosis...). Together with the Monte Carlo method or the method of moments, the stochastic collocation and non-intrusive polynomial chaos expansion methods introduced in the previous chapter “[General Introduction to Polynomial Chaos and Collocation Methods](#)” are widely used for evaluating stochastic objective functions. We more particularly focus on the latter approach in this chapter.

The polynomial chaos (PC) or homogeneous chaos expansion defined as the span of Hermite polynomial functionals of a Gaussian random variable has been introduced by Wiener [1] for stochastic processes. Mean-square convergence is guar-

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anted by the Cameron-Martin theorem [2] and is optimal (i.e., exponential) for Gaussian processes. For arbitrary random processes, the numerical study in [3] has shown that the convergence rates are not optimal. This observation has prompted the development of generalized chaos expansions (gPC) involving other families of polynomials [3, 4]. They consist in expanding any function of random variables into a linear combination of orthogonal polynomials with respect to the probability density functions (PDFs) of these underlying random variables. PC and gPC expansions have recently received a broad attention in engineering sciences, where they are extensively used as a constructive tool for representing random vectors, matrices, tensors, or fields for the purpose of quantifying uncertainty in complex systems. Several implementation issues and applications are described in, e.g., [5–13] and references therein.

The intrusive PC expansion originally introduced in [7, 14, 15] is based on a Galerkin-type projection formulation of the model equations, typically the incompressible or compressible Navier–Stokes equations in CFD, to derive the governing equations for the spectral expansion coefficients of the output quantities of interest. More precisely, the PC expansions of the model parameters and variables are substituted in the model equations, which in turn yield the evolution equations for the outputs from Galerkin projections using the orthogonal polynomials of the PC expansions [6, 7, 10]. The projection coefficients are thus obtained by solving ordinary differential equations in time. Regarding non-intrusive PC expansions, two approaches for computing the PC coefficients of the output quantities of interest have usually been considered: (i) the projection approach, in which they are computed by structured quadratures, i.e., Gauss quadratures, or unstructured quadratures, i.e., Monte Carlo or quasi Monte Carlo sampling; and (ii) the regression approach, minimizing some error measure or truncation tolerance of the PC expansion for some particular values of the inputs (which can be the quadrature sets invoked just above, for example). Both techniques suffer from some well-identified shortcomings when the dimension of the parameter space, and the number of model evaluations alike, increases. Indeed, a PC expansion of total degree p in N variable parameters contains $P = \binom{p+N}{N}$ coefficients. A direct way to compute them is to use a tensor product grid in the parameter space requiring about $q \sim \left(\frac{p}{2}\right)^N$ evaluations of the process. These q runs are very often unaffordable for large parameter spaces and complex configurations, as in CFD for example. Fortunately, the Smolyak algorithm [16] defines sparse grid quadratures involving $q \sim \left(\frac{p}{2}\right)^{\log N}$ points while preserving a satisfactory level of accuracy. In [17], it has been observed that such sparse rules typically become competitive with respect to tensor grids for dimensions $N \geq 4$. Consequently, collocation techniques with sparse quadratures or adaptive regression strategies have been developed in order to circumvent the dimensionality concern [18–25].

In the application presented in this chapter, we adopt the regression approach. We also aim at benefiting from the sparsity of the process outputs themselves to reconstruct their PC representations in a non-adaptive way [26]. Indeed, we rely on the common observation that many cross-interactions between the input parameters are actually smoothened, or even negligible, once that have been propagated to

some global quantities of interest processed from complex aerodynamic computations. The corresponding PC expansions should thus involve only low-order polynomials, such that the contribution of the higher-order polynomials is negligible. We can therefore expect to achieve a successful output recovery by the techniques known under the terminology of compressed sensing [27, 28]. In this theory, the reconstruction of a sparse signal on a given, known basis requires only a limited number of evaluations at randomly selected points—at least significantly less than the a priori dimension P of the basis. We thus resort to unstructured sampling sets to recover sparse outputs. Compressed sensing is formulated as a constrained, underdetermined system which may be solved by convex optimization algorithms. The rest of this chapter is organized as follows. The formal framework of non-intrusive UQ is briefly presented in the next section. The gPC expansion method itself is addressed in section “[Generalized Polynomial Chaos Expansion](#),” which is more particularly focused on the non-intrusive computation of the PC coefficients by either projection or regression approaches. This framework is illustrated in section “[Application to Transonic Turbulent Flow Around a Two-Dimensional RAE 2822 Airfoil](#)” where the application of the foregoing techniques to a two-dimensional transonic turbulent flow around a RAE 2822 airfoil modeled by steady-state Reynolds-averaged Navier–Stokes (RANS) equations is considered. The efficiency of a sparsity-based reconstruction approach is emphasized. Some conclusions are finally drawn in section “[Conclusions](#)”.

Uncertainty Quantification and Propagation: Model Problem

In CFD applications, numerical models are built to simulate complex fluid flows around rigid or flexible profiles. They are implemented in computer programs which tend to become more and more sophisticated and extensive. These models often exhibit certain features that may be considered as uncertain, or they depend on parameters that may be considered as such. Propagation and quantification of uncertainty aim at establishing a quantitative assessment of and some insight into the impact that these uncertainties have on the predictions given by the models. Statements about the influence of uncertainties may serve to guide the allocation of resources in order to reduce them, but are also essential in the process of validating the models in the presence of uncertain parameters or in the process of optimizing a design, among other objectives [29].

In this chapter, we think of a computational model as a linear or nonlinear mapping F of a set of input parameters ξ in $\Gamma \subseteq \mathbb{R}^N$ into a quantity of interest y ; that is,

$$y = F(\xi), \quad F : \mathbb{R}^N \rightarrow \mathbb{R}, \quad (1)$$

assuming without loss of generality that the quantity of interest is a scalar. Vector-valued quantities of interest may be considered alike, following the same lines as

exposed below. Here, it is also implicitly assumed that the uncertainties affecting the model are embedded in the input variables ξ , or a subset of these variables; that is, modeling uncertainties are not accounted for. In other words, modeling errors arising from modeling simplifications and assumptions are not considered here. Output-predictor uncertainties pertaining to the quantity of interest, which may be subject to an additive random noise, are also disregarded in this setting. Therefore, in the so-called parametric approach promoted in this chapter, the uncertain input variables ξ are representative of variable geometrical characteristics, boundary conditions, loadings, physical or mechanical properties, or combinations of them. These input parameters are represented by a $\mathbb{R}^N$ -valued random variable $\Xi = (\Xi_1, \Xi_2, \dots, \Xi_N)$, the characterization of which requires the knowledge of its probability distribution $\mathcal{P}_\Xi$. Ξ can be a continuous or a discrete, or even a combination of both, random variable. In the former case, it is understood that the probability distribution thus admits a probability density function (PDF) $\xi \mapsto W_\Xi(\xi)$ with values in $\mathbb{R}_+ = [0, +\infty]$ such that $\mathcal{P}_\Xi(\mathcal{B}) = \int_{\mathcal{B}} W_\Xi(\xi) d\xi$ for any subset $\mathcal{B}$ of $\mathbb{R}^N$.

The mapping of Ξ through the computational model F then provides the characterization of the quantity of interest as a real-valued random variable Y such that $Y = F(\Xi)$. By the causality principle, the probability distribution $\mathcal{P}_Y$ of Y is the probability distribution of Ξ transported by the model F , i.e.:

$$\mathcal{P}_Y(\mathcal{B}) = \mathcal{P}_\Xi(\xi \in \mathbb{R}^N; F(\xi) \in \mathcal{B}). \quad (2)$$

The subsequent computation of statistical descriptors of the quantity of interest is straightforward. Considering for example the mean μ_Y and variance σ_Y^2 , one has:

$$\begin{aligned} \mu_Y &= \int_{\mathbb{R}} y \mathcal{P}_Y(dy) = \int_{\mathbb{R}^N} F(\xi) \mathcal{P}_\Xi(d\xi), \\ \sigma_Y^2 &= \int_{\mathbb{R}} (y - \mu_Y)^2 \mathcal{P}_Y(dy) = \int_{\mathbb{R}^N} (F(\xi) - \mu_Y)^2 \mathcal{P}_\Xi(d\xi), \end{aligned} \quad (3)$$

assuming these integrals remain bounded; i.e., Y has finite variance; otherwise, the statistical descriptors do not exist.

The evaluation of the multi-dimensional integrals above requires a thorough knowledge of the computational model F , which of course has no analytical expression for complex configurations of industrial relevance. This can be achieved intrusively, by expliciting all dependences of the underlying physical and computational models with respect to the parameters; see, e.g., [7, 11, 14, 15, 30, 31] and references therein. This approach is seldom considered in CFD because the nonlinearity of the physical model and complexity of the codes makes the implementation tricky. Alternatively, the non-intrusive approach consists in expressing all dependences of the quantity of interest Y with respect to the parameters by carefully chosen sampling procedures or surrogates (also called response surfaces); see, e.g., [17–20, 32–39]. This is the approach retained in the following. More particularly, we address the construction of polynomial surrogates using a family of multivariate orthogonal polynomials with respect to the probability distribution $\mathcal{P}_\Xi$ of the random input parameters.

Generalized Polynomial Chaos Expansion

Polynomial Surrogates

Let $\mathbf{j} = (j_1, j_2, \dots, j_N) \in \mathbb{N}^N$ be a so-called multi-index, we denote by $\xi^{\mathbf{j}}$ the multivariate monomial associated to that multi-index as a function from $\mathbb{R}^N$ into $\mathbb{R}$ such that $\xi^{\mathbf{j}} = \xi_1^{j_1} \times \xi_2^{j_2} \times \dots \times \xi_N^{j_N}$. The modulus of $\mathbf{j}$, namely $|\mathbf{j}| = j_1 + j_2 + \dots + j_N$, is also the total order of the monomial $\xi^{\mathbf{j}}$. Then a polynomial surrogate model G^p of order p is defined as an N -variate polynomial approximating the computational model F as precisely as possible in the $\mathcal{P}_{\Xi}$ -weighted least-squares sense, i.e.:

$$F \simeq G^p = \sum_{|\mathbf{j}|=0}^P c_{\mathbf{j}} \xi^{\mathbf{j}}, \quad \mathbf{c} = \arg \min_{\mathbf{d} \in \mathbb{R}^P} \frac{1}{2} \int_{\mathbb{R}^N} \left| F(\xi) - \sum_{|\mathbf{j}|=0}^P d_{\mathbf{j}} \xi^{\mathbf{j}} \right|^2 \mathcal{P}_{\Xi}(d\xi), \quad (4)$$

where P is the number of monomials $\xi^{\mathbf{j}}$ such that $0 \leq |\mathbf{j}| \leq p$. Conditions guaranteeing the existence (and possible uniqueness) of a solution to the above problem and convergence of the surrogate G^p as its order p is increased are summarized in [29]. In particular, it should be emphasized that provided the surrogate G^p converges to the computational model in the $\mathcal{P}_{\Xi}$ -mean-square sense, that is,

$$\lim_{p \rightarrow +\infty} \int_{\mathbb{R}^N} |F(\xi) - G^p(\xi)|^2 \mathcal{P}_{\Xi}(d\xi) = 0,$$

the probability distribution $\mathcal{P}_{Y^p}$ of the random variable $Y^p = G^p(\Xi)$ converges to the probability distribution $\mathcal{P}_Y$ of Y , Eq. (2), as $p \rightarrow +\infty$.

Polynomial Chaos Surrogates

Consider now $\mathcal{B}^p = \{\psi_{\mathbf{j}}; 0 \leq |\mathbf{j}| \leq p\}$ a set of N -variate polynomials $\psi_{\mathbf{j}}$ spanning the set of all polynomials of total order at most p and orthonormal with respect to $\mathcal{P}_{\Xi}$, that is,

$$\int_{\mathbb{R}^N} \psi_{\mathbf{j}}(\xi) \psi_{\mathbf{k}}(\xi) \mathcal{P}_{\Xi}(d\xi) := \langle \psi_{\mathbf{j}}, \psi_{\mathbf{k}} \rangle = \delta_{\mathbf{j}\mathbf{k}}, \quad 0 \leq |\mathbf{j}|, |\mathbf{k}| \leq p, \quad (5)$$

where $\delta_{\mathbf{j}\mathbf{k}}$ is the N -dimensional Kronecker symbol such that $\delta_{\mathbf{j}\mathbf{k}} = 1$ if $\mathbf{j} = \mathbf{k}$ and $\delta_{\mathbf{j}\mathbf{k}} = 0$ otherwise. The existence of this basis is guaranteed provided that all monomials $\xi^{\mathbf{j}}$, $0 \leq |\mathbf{j}| \leq p$, are $\mathcal{P}_{\Xi}$ -square integrable and the $\mathcal{P}_{\Xi}$ -mean-square norm of all non null polynomials of total order p is nonvanishing. Then, a polynomial surrogate G^p may be constructed equivalently to the $\mathcal{P}_{\Xi}$ -weighted least-squares problem (4) by:

$$F \simeq G^p = \sum_{|\mathbf{j}|=0}^p G_{\mathbf{j}} \psi_{\mathbf{j}}, \quad \mathbf{G} = \arg \min_{\mathbf{g} \in \mathbb{R}^p} \frac{1}{2} \int_{\mathbb{R}^N} \left| F(\boldsymbol{\xi}) - \sum_{|\mathbf{j}|=0}^p g_{\mathbf{j}} \psi_{\mathbf{j}}(\boldsymbol{\xi}) \right|^2 \mathcal{P}_{\boldsymbol{\Xi}}(d\boldsymbol{\xi}), \quad (6)$$

where $\mathbf{G} = \{G_{\mathbf{j}}; 0 \leq |\mathbf{j}| \leq p\}$ and P is again the number of polynomials in the basis $\mathcal{B}^p$. Owing to the orthonormality of the polynomials of that basis, the solution of this problem reads:

$$G_{\mathbf{j}} = \langle F, \psi_{\mathbf{j}} \rangle, \quad 0 \leq |\mathbf{j}| \leq p. \quad (7)$$

The random variable $Y^p = \sum_{|\mathbf{j}|=0}^p G_{\mathbf{j}} \psi_{\mathbf{j}}(\boldsymbol{\Xi})$ is known as a polynomial chaos expansion of the quantity of interest Y . The basis $\mathcal{B}^p$ is referred to as the polynomial chaos (PC), or homogeneous chaos in the original terminology of Wiener [1]. Here, the random parameters $\boldsymbol{\Xi}$ were Gaussian variables; the extension to non-Gaussian variables in, e.g., [3, 4] has been referred to as generalized polynomial chaos (gPC). Lastly, the vector $\mathbf{G}$ is formed with the so-called polynomial chaos coefficients which completely characterize the surrogate model G^p since the polynomial chaos is known from the probability measure $\mathcal{P}_{\boldsymbol{\Xi}}$ of the random inputs. Several methods have been developed to compute them, which are usually classified into intrusive or non-intrusive approaches; see the introductory chapter “[General Introduction to Polynomial Chaos and Collocation Methods](#)”.

Computation of Polynomial Chaos Coefficients

The original intrusive approach introduced in [7, 14, 15] is based on a Galerkin-type projection formulation of the model equations (typically partial differential equations, PDEs) onto a prescribed basis of orthonormal polynomials. The procedure results in a so-called spectral problem formulated in terms of the polynomial chaos coefficients of the solution of the model PDEs, which often requires important modifications of the associated computational model. For this reason, intrusive methods are seldom considered in CFD applications as already noticed above. On the other hand, non-intrusive approaches do not require any modification of the computational model. The non-intrusive projection method numerically determines the polynomial surrogate by approximating the integrals in (7) using dedicated structured (typically Gauss integration rules) or unstructured (e.g., Monte Carlo or quasi Monte Carlo methods) sets of samples of the random input parameters $\boldsymbol{\Xi}$. The interpolatory collocation method determines the polynomial surrogate by interpolating between a set of evaluations of the computational model. It is strongly related to the projection method as outlined in chapter “[General Introduction to Polynomial Chaos and Collocation Methods](#)”; see also the discussion in, e.g., [40]. In fact, it is also called the pseudospectral collocation method in [13, 41]. Introducing a quadrature rule of integration $\boldsymbol{\Theta}(N, Q) = \{\boldsymbol{\xi}^l, w^l; 1 \leq l \leq Q\}$ formed by Q nodes $\boldsymbol{\xi}^l$ in $\mathbb{R}^N$ and associated weights w^l in $\mathbb{R}$, the integral of any smooth, integrable function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ with respect to $\mathcal{P}_{\boldsymbol{\Xi}}$ can be approximated by the weighted sum:

$$\int_{\mathbb{R}^N} f(\boldsymbol{\xi}) \mathcal{P}_{\Xi}(d\boldsymbol{\xi}) \simeq \sum_{l=1}^Q w^l f(\boldsymbol{\xi}^l).$$

Applying this rule to the computation of the polynomial chaos coefficients of Eq. (7), one arrives at the following surrogate model $G^{p,Q}$:

$$F \simeq G^{p,Q} = \sum_{|\mathbf{j}|=0}^p G_{\mathbf{j}}^Q \psi_{\mathbf{j}}, \quad G_{\mathbf{j}}^Q = \sum_{l=1}^Q w^l F(\boldsymbol{\xi}^l) \psi_{\mathbf{j}}(\boldsymbol{\xi}^l), \quad 0 \leq |\mathbf{j}| \leq p. \quad (8)$$

The families of orthonormal polynomials forming $\mathcal{B}^p$ and associated Gauss quadrature rules are explicitly known and tabulated for some labeled probability distributions $\mathcal{P}_{\Xi}$, such as Gamma distributions (corresponding to Laguerre polynomials), Beta distributions (corresponding to Jacobi polynomials, including Legendre polynomials for uniform distributions), Gaussian distributions (corresponding to Hermite polynomials), Poisson distributions (corresponding to Charlier polynomials), binomial distributions (corresponding to Krawtchouk polynomials); see [4, 11, 13, 30]. Alternatively, they can be generated numerically using the Gramm-Schmidt, Stieltjes or Chebyshev algorithm [42] and the Golub-Welsh algorithm [43] to compute their associated Gauss nodes and weights. This procedure has been applied in, e.g., [44–46]. Another possibility is to consider unstructured quadrature sets, whereby the nodes are sampled randomly or quasi-randomly in the domain of integration $\Gamma \subseteq \mathbb{R}^N$ according to their distribution $\mathcal{P}_{\Xi}$, and the associated weights are typically $w^l = \frac{1}{Q}$: This is the Monte Carlo method and its by-products.

Another approach to determine the polynomial chaos coefficients is linear regression. In this setting, the set $\boldsymbol{\Theta}(N, Q)$ is used to form a linear system in the unknowns $\mathbf{G}$ by simply evaluating the surrogate model G^p at the nodes $\{\boldsymbol{\xi}^l; 1 \leq l \leq Q\}$:

$$\boldsymbol{\Phi} \mathbf{G} = \mathbf{y}, \quad (9)$$

where $\mathbf{y} = \{y^l = F(\boldsymbol{\xi}^l); 1 \leq l \leq Q\}$ is the vector of observations of the computational model F at the sampling nodes, and $[\boldsymbol{\Phi}]_{lj} = \psi_{\mathbf{j}}(\boldsymbol{\xi}^l)$ is the so-called $Q \times P$ **measurement matrix**. Numerous methods are available to solve this problem whenever $Q \geq P$, for example least-squares minimization [37, 47]. Likewise different strategies exist for the choice of the sampling nodes $\{\boldsymbol{\xi}^l; 1 \leq l \leq Q\}$ which are reviewed in, e.g., [29, 48–50]. We do not follow this approach in the subsequent developments though. We are rather interested in the situation whereby $Q \leq P$ and more particularly $Q \ll P$, that is, underdetermined systems. This can be achieved thanks to some recent mathematical results pertaining to the resolution of under-sampled linear systems promoting sparsity of the sought solution, known as compressed sensing or compressive sampling [27, 28]. A recent review of the application of this approach to polynomial surrogates reconstruction is proposed in [51]; see also [17, 26, 52–61]. The compressed sensing approach consists in reformulating Eq. (9) considered as an underdetermined system as a minimization problem with some sparsity constraint, namely:

$$\mathbf{G}^* = \arg \min_{\mathbf{g} \in \mathbb{R}^P} \{ \|\mathbf{g}\|_m; \mathbf{y} = \Phi \mathbf{g} \}, \quad (P_{m,0})$$

with the ℓ_m -norm $\|\mathbf{g}\|_m = (\sum_{j=0}^{P-1} |g_j|^m)^{\frac{1}{m}}$, $m > 0$, and $\|\mathbf{g}\|_0 = \#\{j; g_j \neq 0\}$ otherwise. **Sparsity** means that only a small fraction of the sought coefficients $\mathbf{G}^*$ are nonzero. Such a constraint is added to ensure well-posedness of (9) when $Q < P$ by seeking a solution with the minimum number of nonzero terms. The case $m = 0$ however yields a non-unique solution in general, which is in addition NP hard to compute as the cost of global search is exponential in P . Further researches in compressed sensing have shown that the convex relaxation of $(P_{0,0})$ by considering the ℓ_1 -norm instead yields a unique solution provided that the latter is sufficiently sparse and that the measurement matrix Φ has some prescribed properties. The problem $(P_{1,0})$ is referred to as basis pursuit [62] in the dedicated literature.

Now, the equality (9) is often too restrictive because the truncated p -th order polynomial chaos basis $\mathcal{B}^p$ is not complete for the exact representation of the observations $\mathbf{y}$. Thus, a truncation error has also to be accounted for in the solution process. This is accommodated by reformulating $(P_{1,0})$ as:

$$\mathbf{G}^* = \arg \min_{\mathbf{g} \in \mathbb{R}^P} \{ \|\mathbf{g}\|_1; \|\mathbf{y} - \Phi \mathbf{g}\|_2 \leq \varepsilon \}, \quad (P_{1,\varepsilon})$$

for some tolerance $0 \leq \varepsilon \ll 1$ on the polynomial chaos truncation. The latter problem is known as basis pursuit denoising (BPDN) [62]. The successful recovery of $\mathbf{G}$ of Eq. (9) by solving $(P_{1,\varepsilon})$ is guaranteed by the **restricted isometry property** (RIP) which the measurement matrix Φ has to satisfy. For each integer $S \in \mathbb{N}^*$, the isometry constant δ_S of Φ is defined as the smallest number such that:

$$(1 - \delta_S) \|\mathbf{g}_S\|_2^2 \leq \|\Phi \mathbf{g}_S\|_2^2 \leq (1 + \delta_S) \|\mathbf{g}_S\|_2^2$$

for all S -sparse vectors $\mathbf{g}_S \in \{\mathbf{g} \in \mathbb{R}^P; \|\mathbf{g}\|_0 \leq S\}$. Then, Φ is said to satisfy the RIP of order S if, say, δ_S is not too close to 1. This property amounts to saying that all S -column submatrices of Φ are numerically well-conditioned, or S (or less) columns selected arbitrarily in Φ are nearly orthogonal. Consequently, they form a near isometry so that Φ approximately preserves the Euclidean norm of S -sparse vectors. The following theorem by Candès et al. [27, Theorem 1.2], [63, Theorem 3] then states that $(P_{1,\varepsilon})$ can be solved efficiently:

Theorem 1 ([27, 63]) *Assume $\delta_{2S} < \sqrt{2} - 1$. Then, the solution $\mathbf{G}^*$ to $(P_{1,\varepsilon})$ satisfies:*

$$\|\mathbf{G}^* - \mathbf{G}\|_2 \leq C_0 \frac{\|\mathbf{G}_S - \mathbf{G}\|_1}{\sqrt{S}} + C_1 \varepsilon$$

for some $C_0, C_1 > 0$ depending only on δ_{2S} .

This result calls for several comments. First, the coefficients $\mathbf{G}$ are actually nearly sparse, rather than strictly sparse, in the sense that only a small fraction of them contribute significantly to the output statistics while the others are not strictly null.

Opportunately, the foregoing theorem deals with all signals and not only the S -sparse ones. Second, it also deals with noiseless recovery if $\varepsilon = 0$. Third, it is deterministic and does not involve any probability for a successful recovery. Lastly, the ℓ_1 -minimization strategy is non-adapted because it identifies the sparsity pattern, that is the order (location) of the negligible coefficients in the polynomial chaos basis, and the leading coefficients at the same time. The algorithm can therefore efficiently capture the relevant information of a sparse vector without trying to comprehend that vector [63]. This is clearly a much desirable feature for practical industrial applications.

Application to Transonic Turbulent Flow Around a Two-Dimensional RAE 2822 Airfoil

The foregoing strategy of using a gPC expansion for uncertainty propagation and quantification is applied to an aerodynamic problem taken from [17]. Here, we consider a two-dimensional transonic turbulent flow around a RAE 2822 airfoil modeled by the steady-state Reynolds-averaged Navier Stokes (RANS) equations together with a Spalart-Allmaras turbulence model closure [64]. The nominal flow conditions are the ones described in Cook et al. [65] for the test case #6 together with the wall interference correction formulas derived in [66, pp. 386–387] and their slight modifications suggested in [67, p. 130] (see also the CFD verification and validation Web site of the NPARC Alliance [68]). The nominal free-stream Mach number $\underline{M}_\infty = 0.729$, angle of attack $\underline{\alpha}_\infty = 2.31^\circ$, and Reynolds number $\underline{Re} = 6.50 \cdot 10^6$ (based on the airfoil chord length c , fluid velocity, temperature, and molecular viscosity at infinity) arise from the corrections $\Delta M_\infty = 0.004$ and $\Delta \alpha_\infty = -0.61^\circ$ given in [67, p. 130] for the test case #6 outlined in Cook et al. [65], for which $\underline{M}_\infty = 0.725$, $\underline{\alpha}_\infty = 2.92^\circ$, and $\underline{Re} = 6.50 \cdot 10^6$. At last, the far-field temperature is fixed at $\underline{T}_\infty = 300$ K and the ratio of specific heats of the air at $\gamma = 1.4$.

Definition of the Uncertainties

Our aim is to characterize the influence of variabilities of the free-stream Mach number M_∞ and angle of attack α_∞ (operational parameters), and of the thickness-to-chord ratio $r = h/c$ (geometrical parameter) on some aerodynamic quantities of interest, such as the drag, lift, or pitching moment coefficients C_D , C_L , or C_m , respectively. These variable parameters are assumed to be independent and to follow Beta laws of the first kind β_1 . Therefore, their marginal PDFs read:

$$\beta_1(\xi; a, b) = \mathbb{1}_{[X_m, X_M]}(\xi) \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{(\xi - X_m)^{a-1} (X_M - \xi)^{b-1}}{(X_M - X_m)^{a+b-1}}.$$

Table 1 Symmetric β_1 laws for the variable geometrical and operational parameters

	$a = b$	X_m	X_M
Ξ_1	4	$0.97 \times \underline{r}$	$1.03 \times \underline{r}$
Ξ_2	4	$0.95 \times \underline{M}_\infty$	$1.05 \times \underline{M}_\infty$
Ξ_3	4	$0.98 \times \underline{\alpha}_\infty$	$1.02 \times \underline{\alpha}_\infty$

In the above, $\Gamma(z) = \int_0^{+\infty} t^{z-1} e^{-t} dt$ is the usual Gamma function, and $[X_m, X_M]$ is the compact support of the random parameter $\Xi \sim \beta_1$. The parameters $a = b$ as well as the bounds X_m, X_M for the three variable parameters $\xi_1 = r$, $\xi_2 = M_\infty$, $\xi_3 = \alpha_\infty$ are gathered in the Table 1. This definition of uncertainties is part of the FP7 UMRIDA Project (<http://www.umrida.eu>), which gathers a novel data base of industrial challenges with prescribed uncertainties for the validation of UQ techniques against this series of relevant industrial test cases. We note in passing that the β_1 model is the one arising from Jaynes' maximum entropy principle [69] when constraints on (i) the boundedness of the support $[X_m, X_M]$, and (ii) the values of the averages $\mathbb{E}\{\log(\Xi - X_m)\}$ and $\mathbb{E}\{\log(X_M - \Xi)\}$, are imposed.

Polynomial Basis and Sampling Nodes

From the analysis of section “Generalized Polynomial Chaos Expansion,” it is seen that the main ingredients requested for the construction of polynomial surrogates of the quantity of interest $y = C_D$, C_L , or C_m are the truncated polynomial basis $\mathcal{B}^p$ and the quadrature rule $\Theta(N, Q)$, for Q integration nodes and a total number of polynomials $P = \binom{N+p}{N}$. In addition, we have here $N = 3$ for the parameter space dimension. Owing to the choices made for the variable parameters considered for this case (see Table 1), we have $\xi = (\xi_1, \xi_2, \xi_3) \in \Gamma = \prod_{j=1}^3 [X_m^{(j)}, X_M^{(j)}]$ and $\mathcal{P}_\Xi(\xi) = \bigotimes_{j=1}^3 \beta_1(\xi_j; 4, 4)$. Therefore, the integration nodes should be chosen from a Gauss-Jacobi quadrature rule, and the polynomial basis $\mathcal{B}^p$ should be constituted by the multivariate Jacobi polynomials which are orthogonal with respect to the weight function $\xi \mapsto W_3(\xi) = \prod_{j=1}^3 W_1(\xi_j)$ on $[-1, 1]^3$ (after a proper renormalization of Γ), where $\xi \mapsto W_1(\xi_j) = (1 - \xi_j^2)^3$. They are computed by:

$$\psi_{\mathbf{j}}(\xi) = \prod_{k=1}^3 \psi_{j_k}(\xi_k), \quad |\mathbf{j}| \leq p,$$

where $\{\psi_{j_k}; j_k \geq 0\}$ is the family of one-dimensional orthonormal Jacobi polynomials with respect to the weight function W_1 . In the present study, the polynomial surrogates G^p constructed for the evaluation of the drag, lift, and pitching moment coefficients are truncated up to the total order $p = 8$. Therefore, $P = \binom{p+3}{3} = 165$ multivariate Jacobi polynomials are ultimately retained in those gPC expansions.

As for the sampling nodes $\{\xi^l; 1 \leq l \leq Q\}$, one may choose them as (i) the nodes of a quadrature rule $\Theta(N, Q)$ so that Eq. (8) can be used; or (ii) select them randomly according to their PDF $\mathcal{P}_{\Xi}$ so that $(P_{1,\epsilon})$ can be used. Indeed, selecting the sampling nodes randomly eases the fulfillment of the RIP and consequently increases the efficiency of ℓ_1 -minimization. The use of a structured sampling set (quadrature rule) in the latter procedure is examined further on in [58], though. Quadrature rules for the approximation of multivariate integrals have been introduced in the chapter “General Introduction to Polynomial Chaos and Collocation Methods” of this book. Univariate Gauss-Jacobi-Lobatto (GJL) quadratures $\Theta(1, q)$ are aimed at integrating a smooth function $\xi \mapsto f(\xi)$ defined on $[-1, 1]$ by:

$$\int_{-1}^1 f(\xi)(1-\xi)^{a-1}(1+\xi)^{b-1}d\xi \simeq \sum_{l=1}^{q-2} w^l f(\xi^l) + w^{q-1}f(-1) + w^q f(1), \quad (10)$$

where $a, b > -1$, such that this rule turns to be exact for polynomials up to the order $2q - 3$. Here, the bounds ± 1 are explicitly included in the quadrature nodes. The reason why we include them in the rule stems from the fact that the basic engineering practice would typically consider the evaluation of the physical model F at the bounds of the support of the variable parameters. The main advantage of using Gauss-Jacobi quadratures is that they do not add integration nodes for the increased order $a + b - 2$ induced by the weight function $(1 - \xi)^{a-1}(1 + \xi)^{b-1}$. Since in our case, we have chosen a total order $p = 8$, $q = 10$ GJL modes are needed to recover exactly the orthonormality property (5) for the corresponding univariate Jacobi polynomials. Indeed Q should be defined such that $2q - 3 \geq 16$ in this situation.

A multivariate quadrature rule may subsequently be obtained by full or sparse tensorization of the above univariate rule. Firstly, a fully tensorized grid is obtained by the straightforward product rule:

$$\Theta(N, Q) = \bigotimes_{j=1}^N \Theta(1, q), \quad (11)$$

which contains $Q = q^N$ nodes in Γ . Secondly, sparse quadrature rules can be derived thank to the Smolyak algorithm [16]. A brief overview is again given in the chapter “General Introduction to Polynomial Chaos and Collocation Methods” of this book. Such sparse rules have been considered in [17, 19, 22–25] among others but will not be used here, though. Note also that a different number of nodes may be used along each dimension for either product or sparse rules, as proposed in some adaptive strategies [18–21, 23–25]. The fully tensorized rule in $N = 3$ dimensions based on a 10 nodes univariate GJL rule is displayed on Fig. 1(left). It contains $Q = 10^3$ nodes, to be compared with the $Q = 80$ nodes sampled randomly for the application of the ℓ_1 -minimization algorithm $(P_{1,\epsilon})$ (right).

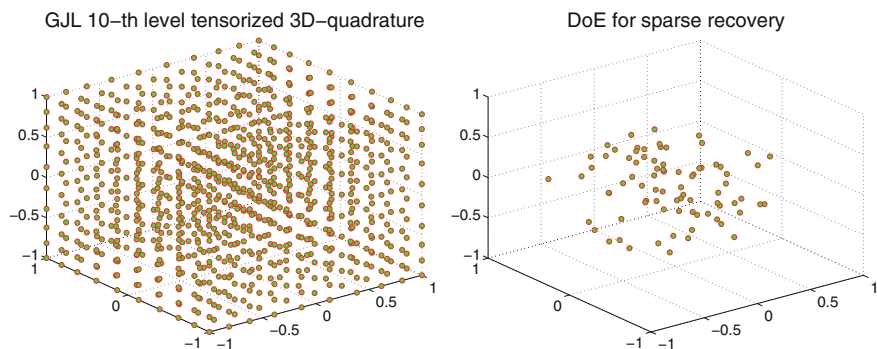


Fig. 1 Left: three-dimensional nodes based on a tensorized univariate GJL rule with $q = 10$ nodes along each dimension and $a = b = 4$ ($Q = 10^3$). Right: randomly sampled nodes following a three-dimensional symmetric β_1 law with $a = b = 4$ ($Q = 80$)

Computational Model

Once the node sets have been defined, either from a fully tensorized quadrature rule or random sampling, it remains to run the computational model for each sampling node ξ^i in those sets to obtain the vector $\mathbf{y}$ of observations of the quantity of interest. The numerical approximation of the aerodynamic problem considered here is computed using the *elsA* software [70]. At first, the RANS equations are discretized in space by a cell-centered finite-volume method. The convective fluxes are computed using the upwind Roe flux and a second-order MUSCL scheme [71] with a van Albada limiter [72]. The diffusive fluxes are computed as the half sum of the normal flux densities in the two adjacent cells sharing the current interface (with due care of the boundary conditions on external interfaces), considering corrected cell-centered gradients of the fluid velocity in these cells. The diffusive fluxes of the transport equation for the turbulent variable in the Spalart-Allmaras model are discretized using a similar approach, whereas first-order Roe fluxes are used for the convective terms. Finally, the source term of the transport equation is computed using the temperature gradients at the center of the cells. Secondly, these semi-discretized RANS equations are solved in time using a backward Euler scheme up to convergence toward a steady-state solution. The linearization of the resulting nonlinear implicit system is performed using the Lower-Upper Symmetric Successive Overrelaxation (LU-SSOR) scheme [73] with four relaxation cycles. The convergence is accelerated by the use of multigrid techniques for steady flows. Uniform flow is considered as the initial conditions for the iterations with respect to the time parameter.

The nominal problem is discretized using a $769c \times 193c$ mesh shown in Fig. 2, where the boundary at infinity was left intentionally far (at about $500c$ from the airfoil). These values proved to be sufficient to avoid spurious reflection with the far-field boundary. Given the large number of simulations to run, the numerical parameters of the steady-state algorithm proved to be essential to insure a fast convergence.

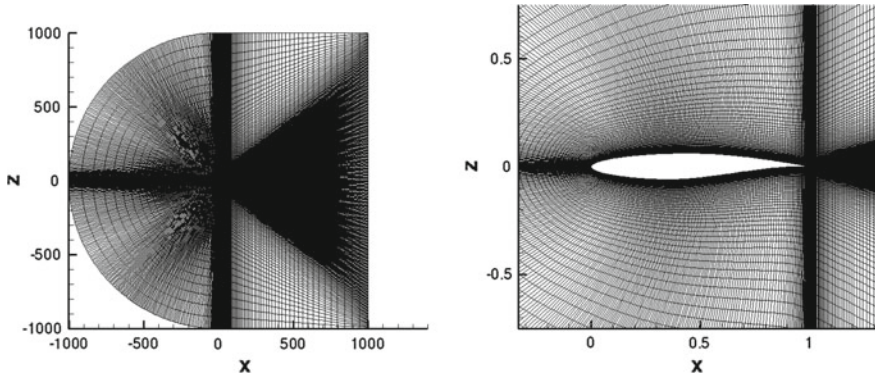


Fig. 2 Computational domain for the baseline configuration: overview of the mesh (left), close view at the airfoil (right)

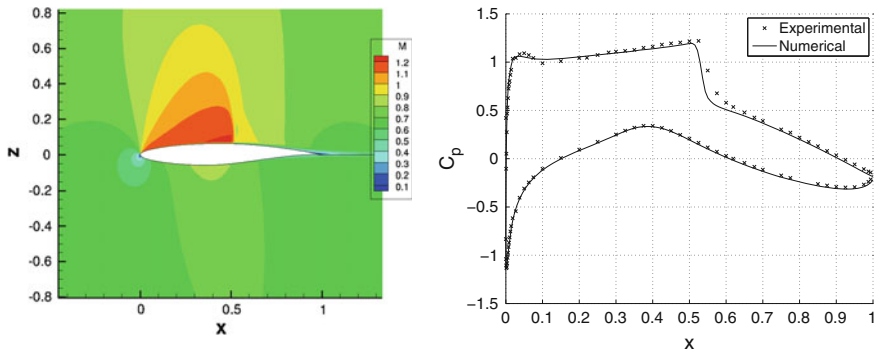


Fig. 3 Magnitude of velocity (left) and static pressure coefficient C_p at the wall (right, solid line) for the baseline configuration $M_\infty = 0.729$, $\alpha = 2.31^\circ$, $Re = 6.50 \cdot 10^6$. The latter is compared with the experiment results gathered on the CFD verification and validation Web site of the NPARC Alliance [68] (crosses)

Once all the foregoing numerical parameters have been fixed, the number of iterations is determined from the evolution of the resulting global forces (not shown here). A number of 2000 iterations appeared to be acceptable, the discrete residuals of all equations and their decrease being checked at every iteration. Hence, this number of iterations has been retained for all subsequent calculations so far. The flow is attached with a weak shockwave on the suction side. The contour plot of the magnitude of the velocity and the static pressure profile at the wall are displayed on Fig. 3 for the baseline configuration $M_\infty = 0.729$, $\alpha_\infty = 2.31^\circ$, and $Re = 6.50 \cdot 10^6$. We see from this latter figure that our numerical results are in good agreement with the experimental results reported in [68].

Results

The mean μ and standard deviation σ of the drag, lift, and pitching moment coefficients C_D , C_L , and C_m , respectively, are gathered in Table 2 for (i) either $Q = 10^3$ calls to the *elsA* computational model F with the 10-th level tensorized GJL rule (see Fig. 1left) to evaluate the polynomial coefficients by projection, Eq. (8); (ii) or $Q = 80$ calls with randomly selected nodes (see Fig. 1right) to evaluate the polynomial coefficients by BPDN ($P_{1,\epsilon}$). The primary reason why we have chosen this sampling size is for its ease of use with the multithreading setup of our CFD software. However, the sparsity of the polynomial surrogates is observed to be $S \simeq 10$ from our numerical results. A common practical observation is that $Q \geq 4S \simeq 40$ or so is usually enough for a successful recovery. Therefore, a new sampling set of $Q = 40$ randomly selected nodes (not displayed here) has been generated and used to construct other surrogate models by ($P_{1,\epsilon}$). The corresponding means and standard deviations are also gathered in Table 2. As for solving this ℓ_1 -minimization problem ($P_{1,\epsilon}$), the Spectral Projected Gradient Algorithm (SPGL1) developed by van den Berg and Friedlander [74] and implemented in the MATLAB package SPGL1 [75], is used. The tolerance was fixed at $\epsilon = 10^{-5}$ and we were able to find a solution for all surrogates with this a priori choice without resorting to cross-validation, for example [26]. Further investigations should be carried on on this topic, though. It should also be noted that no particular sampling strategy, such as stratification, low-discrepancy series, or preconditioning, has been applied at this stage to construct the sampling sets. Moreover, some weighting could be applied to the ℓ_1 -norm as notice in [51] and references therein. We have not considered that possibility either and left it to future works.

The PDFs of the three aerodynamic coefficients considered in this study are displayed on Fig. 4 (in log scale) using the 10-th level tensorized quadrature rule and the 80 samples set. Figure 5 displays the same plots for the 40 samples set. These PDFs were estimated from $N_s = 100,000$ evaluations of the gPC surrogates G^p constructed by either approaches and smoothing out the resulting histograms by a normal kernel density function [76]. The horizontal axes are scaled by the mean value of each coefficient computed with the tensorized quadrature rule. A very good agreement is

Table 2 Mean and variance of the aerodynamic coefficients computed by the 10-th level tensorized GJL rule with $Q = 10^3$ and by ℓ_1 -minimization with $Q = 80$ and $Q = 40$

	GJL product rule ($Q = 10^3$)		ℓ_1 -minimization ($Q = 80$)		ℓ_1 -minimization ($Q = 40$)	
	μ	σ	μ	σ	μ	σ
C_D	133.37e-04	34.13e-04	133.34e-04	34.05e-04	133.21e-04	33.78e-04
C_L	72.27e-02	1.67e-02	72.27e-02	1.67e-02	72.29e-02	1.66e-02
C_m	-453.99e-04	32.24e-04	-453.95e-04	32.18e-04	-453.73e-04	31.98e-04

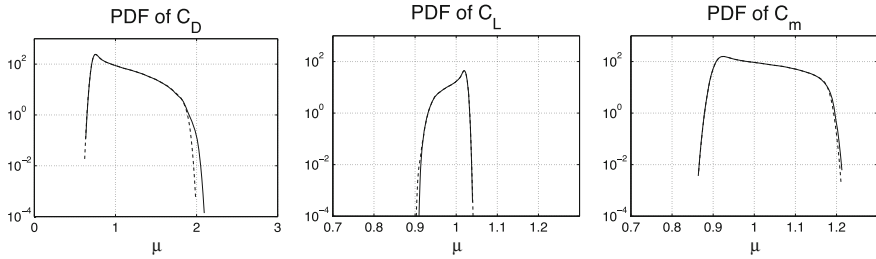


Fig. 4 Comparison of the PDFs of the drag (left), lift (middle) and pitching moment (right) coefficients computed by the 10-th level tensorized GJL rule ($Q = 10^3$, full lines) and ℓ_1 -minimization ($Q = 80$, dashed lines)

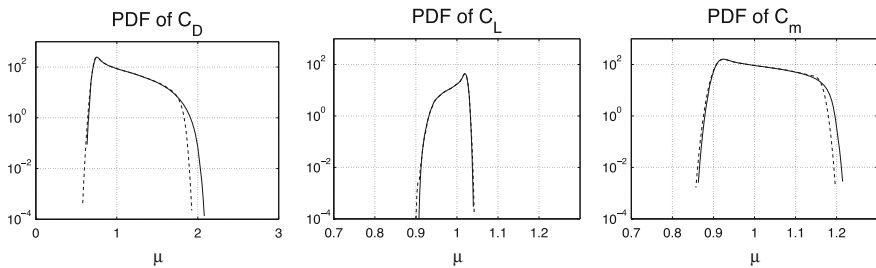


Fig. 5 Comparison of the PDFs of the drag (left), lift (middle) and pitching moment (right) coefficients computed by the 10-th level tensorized GJL rule ($Q = 10^3$, full lines) and ℓ_1 -minimization ($Q = 40$, dashed lines)

achieved by both methods. The reduction of the number of samples to the heuristic lower limit of four times the sparsity S for use in the ℓ_1 -minimization approach essentially affects the tails of the PDFs.

Conclusions

In this chapter we have addressed various methodologies with relevance to the construction of generalized polynomial chaos expansions (gPC) for parameterized complex processes as encountered in computational fluid dynamics. The presentation has been more particularly focused on the use of adapted sampling sets in the parameter space using either structured or unstructured grids to construct the gPC expansions. These techniques were illustrated with the example of a two-dimensional transonic turbulent flow around a RAE 2822 airfoil considering variable geometrical (the thickness-to-chord ratio) and operational (the free-stream Mach number and angle of attack) parameters. The quantities of interest are the usual drag, lift, and pitching moment coefficients for which polynomial surrogates are sought for using the aforementioned sampling sets as learning sets.

Firstly, multivariate quadrature rules based on univariate Gauss-Jacobi-Lobatto rules have been used for the construction of these polynomials surrogates by projection. Secondly, observing a posteriori that the aerodynamic quantities of interest are sparse in the multivariate polynomial chaos basis associated to the parameters probability density functions, an ℓ_1 -minimization approach has been applied in the framework of the theory of compressed sensing. The latter allows to recover the gPC expansion coefficients at a much lower computational cost than the quadrature rules addressed in the first approach. Unstructured sampling nodes are preferred in this process, selecting them randomly in the parameter space. Their number is typically less than the dimension of the polynomial space where the surrogates are sought for, and thus typically much less than the number of nodes of the multivariate quadrature rules that have to be used for a given polynomial order. The ℓ_1 -minimization procedure is non-adaptive in the sense that it identifies both the amplitude of the leading expansion coefficients and their ranks. It thus constitutes a promising direction for future developments in practical applications for more complex geometries and flows, where adaptive strategies within the parametric space, weighted minimization, or preconditioned sampling sets may be advantageous.

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